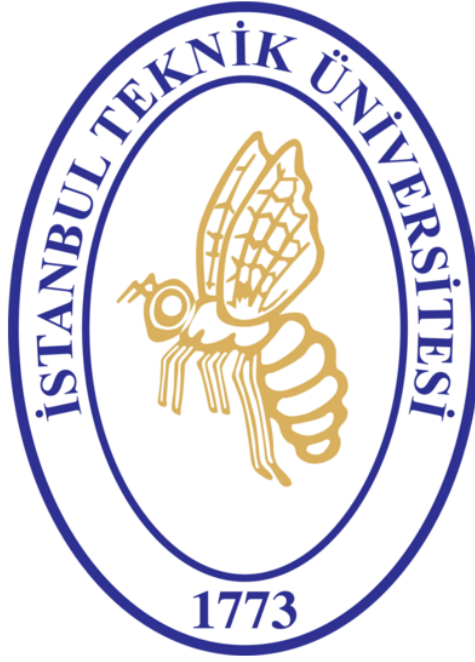


ISTANBUL TECHNICAL UNIVERSITY
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GEOSTATIONARY BENT-PIPE SATELLITE LINK BUDGET
ANALYSIS & ATMOSPHERIC PROPAGATION STUDY



UZH451E – SPACECRAFT COMMUNICATIONS
TERM PROJECT REPORT

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Nomenclature and Abbreviations

Term	Definition
ACM	Adaptive Coding and Modulation
ASI	Adjacent-Satellite Interference
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
BCH	Bose–Chaudhuri–Hocquenghem error-correction code
BPF	Band-Pass Filter
BUC	Block Upconverter
(C/N)	Carrier-to-Noise Ratio
(C/N_0)	Carrier-to-Noise Density Ratio
(C/I)	Carrier-to-Interference Ratio
dB	Decibel
dBHz	Decibel-Hertz
dBW	Decibels referenced to one watt
DTH	Direct-to-Home satellite broadcasting
DVB-S2	Digital Video Broadcasting–Satellite–Second Generation
ECEF	Earth-Centered, Earth-Fixed coordinate frame
EIRP	Equivalent Isotropically Radiated Power
(E_s/N_0)	Symbol Energy to Noise Density Ratio
(E_b/N_0)	Bit Energy to Noise Density Ratio
FEC	Forward Error Correction
FSPL	Free-Space Path Loss
GEO	Geostationary Earth Orbit
GS1	Ground Station 1, Rome uplink station
GS2	Ground Station 2, Ankara downlink station
(G/T)	Antenna Gain-to-System Noise Temperature Ratio
HPA	High-Power Amplifier
HPBW	Half-Power Beamwidth
IF	Intermediate Frequency
IMD	Intermodulation Distortion
ITU-R	International Telecommunication Union Radiocommunication Sector
LDPC	Low-Density Parity-Check code
LNA	Low-Noise Amplifier
LNB	Low-Noise Block Downconverter
MODCOD	Modulation and Coding Scheme
OBO	Output Back-Off
P1dB	One-decibel compression output-power point
QPSK	Quadrature Phase Shift Keying

Term	Definition
RF	Radio Frequency
SNR	Signal-to-Noise Ratio
SSPA	Solid-State Power Amplifier
T_{sys}	System Noise Temperature
TWTA	Traveling-Wave Tube Amplifier
ULPC	Uplink Power Control
VSAT	Very Small Aperture Terminal
XPD	Cross-Polarization Discrimination

1 Introduction

This report evaluates a one-way Ku-band geostationary satellite link from Rome to Ankara. The analysis begins with a conventional clear-sky link budget and then adds interference, atmospheric attenuation, receiver-noise variation, apparent GEO motion, and DVB-S2 adaptive coding and modulation.

1.1 System Architecture and Ku-Band Principles

A circular equatorial orbit at an altitude of approximately 35,786 km has a period close to one sidereal day. A controlled spacecraft in this orbit therefore remains near a fixed longitude as viewed from the ground, allowing fixed parabolic antennas to be used. The corresponding slant range is about 37,000–38,000 km for the sites considered here, which produces more than 200 dB of free-space path loss. Small station-keeping excursions are evaluated separately to determine whether they have a measurable effect on this link.

The payload is modeled as a transparent *bent-pipe* transponder. It filters, frequency-translates, amplifies, and retransmits the received uplink without demodulating the waveform on board.

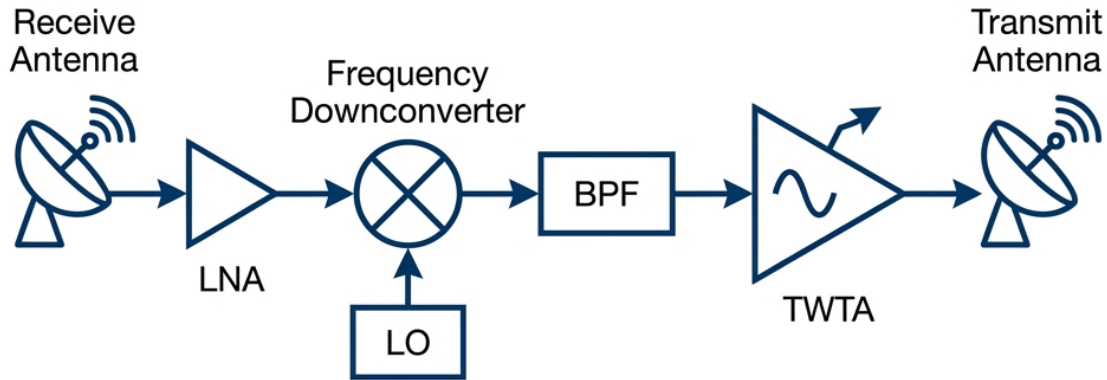


Figure 1: Functional block diagram of a classic transparent bent-pipe satellite transponder payload architecture.

Because the uplink waveform is not regenerated, uplink noise and interference are relayed with the wanted carrier. The uplink and downlink contributions must therefore be combined to obtain the end-to-end $(C/N_0)_{\text{total}}$.

The selected frequencies are 14.0 GHz for the uplink and 12.0 GHz for the downlink. A vertical linear-polarization case is used in the rain model, so ITU-R P.838-3 is evaluated with $\tau = 90^\circ$; a horizontal assignment would require $\tau = 0^\circ$. Ku-band permits useful antenna gain with relatively small apertures, but rain scattering and absorption are significant enough to require an explicit availability analysis.

1.2 Scenario Definition

The baseline is a one-way transparent link from Rome to Ankara through **Eutelsat HOTBIRD 13G**. Figure 2 defines the signal path used throughout the report.



Figure 2: Transparent bent-pipe GEO link architecture between GS1 Rome, Hotbird 13G, and GS2 Ankara.

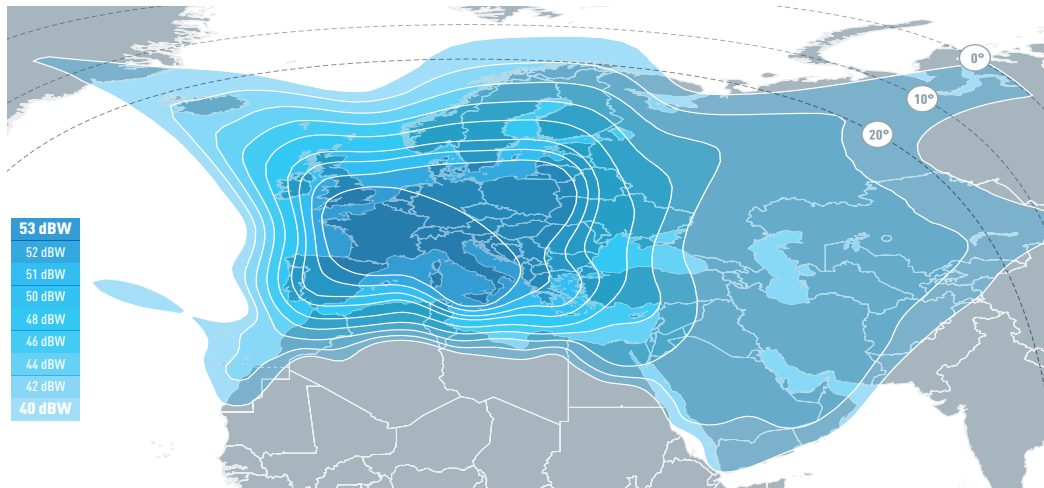
1.2.1 Satellite Selection Rationale

HOTBIRD 13G was selected for this specific Rome–Ankara broadcast case, not as a general ranking of GEO satellites. Its 13°E orbital position gives similar elevation angles at the two terminals, and Eutelsat publishes a Ku-band widebeam EIRP map that can be used for a traceable report-level estimate [1]. Table 2 compares this baseline with Intelsat 39 using the same geometry routine.

Table 2: Candidate satellite comparison for the Rome–Ankara bent-pipe scenario.

Candidate / Operator	Orbital Slot	Rome Geometry	Ankara Geometry	Coverage / Service Context	Selection Decision
Eutelsat HOTBIRD 13G	13°E	41.6° elevation; Rome is close to the orbital longitude.	39.4° elevation.	European and Mediterranean Ku-band DTH and video distribution.	Selected: balanced geometry, an official EIRP map, and a service profile consistent with the modeled DVB-S2 broadcast link.
Intelsat 39 / Intelsat fleet	62°E	20.8° elevation and approximately 39,478 km slant range.	34.9° elevation.	C- and Ku-band broadband, mobility, government, connectivity, and video distribution across Europe, the Middle East, Africa, and Asia.	Not selected: the 62°E slot gives a substantially lower Rome uplink elevation.
Other Intelsat Ku-band spacecraft	Spacecraft-Must be dependent	Must be evaluated for the selected slot and beam.	Must be evaluated for the selected slot and beam.	Potentially suitable for VSAT/IP connectivity or operator-specific leased capacity.	Not used without a formally selected spacecraft, beam, EIRP contour, polarization, transponder plan, and satellite receive G/T.

“Intelsat” denotes an operator fleet rather than one design point. A complete alternative would have to identify the spacecraft, orbital slot, beam, polarization, EIRP contour, transponder plan, and receive G/T. Intelsat 39 at 62°E has suitable regional Ku-band capacity [2, 3], but the Rome elevation is 20.8°, compared with 41.6° for HOTBIRD 13G. The lower elevation increases both slant range and atmospheric path length on the uplink. For the service and sites defined in this report, HOTBIRD 13G therefore provides the more balanced and better-documented baseline.



HOTBIRD 13F & 13G KU-BAND WIDEBEAM DOWNLINK COVERAGE

Figure 3: Official HOTBIRD 13F/13G Ku-band widebeam downlink EIRP contours. The map is reproduced from the Eutelsat HOTBIRD 13 East satellite brochure [1]. A conservative 46 dBW value is used for the Ankara downlink calculation rather than interpolating a higher value from the small-format contour map.



Figure 4: Conceptual illustration of HOTBIRD 13G used to represent the transparent orbital repeater.

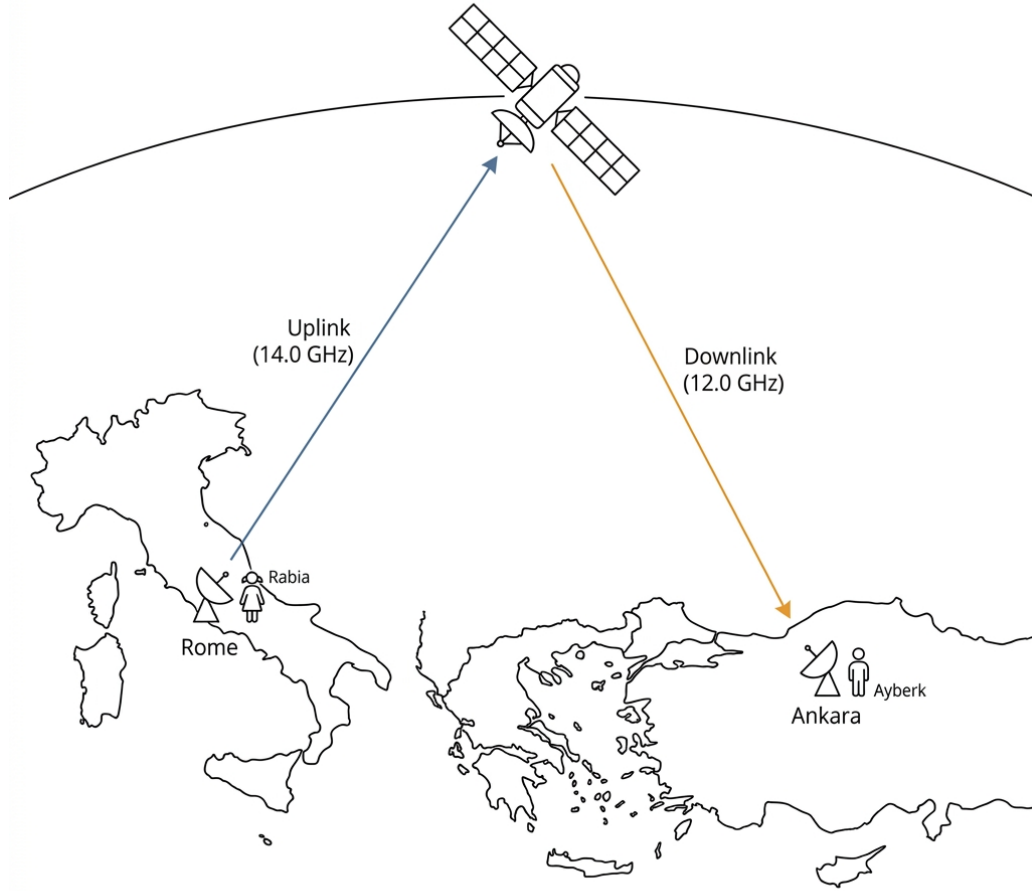


Figure 5: Architecture of the geostationary bent-pipe satellite link connecting GS1 Rome and GS2 Ankara via Eutelsat Hotbird 13G.

- GS1 Rome:** The uplink site is at 41.9028° N, 12.4964° E. The antenna model uses a 2.4 m aperture, consistent with the Andrew Type 243 datasheet, which specifies 48.9 dBi gain at 14.3 GHz [4]. The transmitter is represented by the Norsat ATOMBKU020; its datasheet specifies $P_{\text{dB}} \geq 43$ dBm and a standard 14.0–14.5 GHz transmit range [5]. The link budget uses 20 W as the nominal amplifier output.
- HOTBIRD 13G:** The spacecraft is modeled at 13.0° E. Eutelsat’s published wide-beam map contains 40–53 dBW contours; the Ankara downlink is assigned 46 dBW as a conservative contour-based estimate [1]. The selected transponder receive G/T is not published in the available operator material, so 3.0 dB/K remains an explicit engineering assumption.
- GS2 Ankara:** The receive site is at 39.9334° N, 32.8597° E and 0.94 km altitude. The Triax TD88 sheet specifies a 95 cm by 85 cm offset reflector and 38.8 dBi gain at 11.7 GHz [6]. The calculation represents this aperture by an equivalent 0.9 m circular dish and obtains approximately 39.0 dBi at 12 GHz. The 150 K system noise temperature is an assumption because no LNB-specific noise-temperature measurement is available for the modeled terminal.

1.3 Computational Methodology

The project brief used MATLAB's Satellite Link Budget Analyzer as a reference workflow but allowed an equivalent implementation. The calculations were implemented in Python so that the assumptions, equations, test cases, CSV outputs, and plots could be kept in one reproducible codebase. The clear-sky equations follow Pratt's link-budget procedure [7]; the additional modules implement the stated ITU-R recommendation versions, a separate illustrative Monte Carlo weather model, interference estimates, dynamic receiver noise, apparent GEO motion, and DVB-S2 ACM.

Reproducibility Note

The Python analysis scripts, generated figures, and numerical workflow used in this report are available in the project repository: [GitHub Repository](#).

1.4 Report Structure

Chapter 2 defines the geometry and clear-sky link equations. Chapter 3 describes interference, digital metrics, receiver-noise variation, and GEO motion. Chapter 4 gives the atmospheric and ACM models. Chapter 5 presents the numerical results, validation checks, sensitivity plots, and ablation study; Chapter 6 summarizes the design implications.

2 Link Design Equations

This chapter defines the geometry, antenna, propagation, and noise equations used in the clear-sky calculation. The link-budget sequence follows the procedure presented by Pratt, Bostian, and Allnutt [7]; the same equations are implemented in the Python model.

2.1 Slant-Path Geometry and Coordinate Transformations

Slant range, elevation, and azimuth are calculated by expressing the ground station and satellite positions in an Earth-Centered, Earth-Fixed (ECEF) frame.

From Kepler's third law, the orbital radius r_{sat} for a geosynchronous period T (one sidereal day, approximately 86,164 s) is:

$$r_{sat} = \sqrt[3]{\frac{\mu \cdot T^2}{4\pi^2}} \approx 42164.0 \text{ km} \quad (1)$$

where $\mu \approx 3.986 \times 10^5 \text{ km}^3/\text{s}^2$ is the Earth's standard gravitational parameter. Given the Earth's equatorial radius $R_e = 6378.137 \text{ km}$, the satellite's altitude above the equator is approximately 35786 km.

For a ground station located at latitude ϕ , longitude λ , and altitude h above mean sea level, the ECEF position vector $\vec{R}_{ground}$ is:

$$\vec{R}_{ground} = \begin{bmatrix} (R_e + h) \cos \phi \cos \lambda \\ (R_e + h) \cos \phi \sin \lambda \\ (R_e + h) \sin \phi \end{bmatrix} \quad (2)$$

The geostationary satellite is modeled in the equatorial plane ($\phi_{sat} = 0^\circ$) at longitude λ_{sat} , with Cartesian position vector:

$$\vec{R}_{sat} = \begin{bmatrix} r_{sat} \cos \lambda_{sat} \\ r_{sat} \sin \lambda_{sat} \\ 0 \end{bmatrix} \quad (3)$$

The line-of-sight vector from the ground station to the satellite is:

$$\vec{\rho} = \vec{R}_{sat} - \vec{R}_{ground} \quad (4)$$

The slant range d is the Euclidean norm of this vector:

$$d = \|\vec{\rho}\| = \sqrt{\rho_x^2 + \rho_y^2 + \rho_z^2} \quad (5)$$

The implementation uses a spherical Earth. A geodetic calculation based on the WGS-84 ellipsoid would instead use:

$$x = (N(\phi) + h) \cos \phi \cos \lambda \quad (6)$$

$$y = (N(\phi) + h) \cos \phi \sin \lambda \quad (7)$$

$$z = (N(\phi)(1 - e^2) + h) \sin \phi \quad (8)$$

where $e^2 = 2f - f^2$, $f \approx 1/298.257$, and $N(\phi) = R_e / \sqrt{1 - e^2 \sin^2 \phi}$. At GEO range, the spherical approximation changes the path length by much less than the uncertainty associated with the assumed RF parameters, so it is adequate for this project-level budget.

The elevation angle is obtained by projecting the unit line-of-sight vector $\vec{a}_\rho = \vec{\rho}/d$ onto the local upward unit vector:

$$\vec{u}_{up} = \begin{bmatrix} \cos \phi \cos \lambda \\ \cos \phi \sin \lambda \\ \sin \phi \end{bmatrix} \quad (9)$$

The elevation angle is therefore:

$$\theta = \arcsin(\vec{a}_\rho \cdot \vec{u}_{up}) = \arcsin\left(\frac{\vec{\rho} \cdot \vec{u}_{up}}{d}\right) \quad (10)$$

Azimuth is calculated from the local east and north components and is measured clockwise from true north:

$$Az = \arctan_2(\vec{a}_\rho \cdot \vec{u}_{east}, \vec{a}_\rho \cdot \vec{u}_{north}) \quad (11)$$

2.2 Antenna Parameters and Aperture Efficiency

For a circular parabolic dish antenna of physical diameter D operating at carrier frequency f , the wavelength of the electromagnetic wave is $\lambda_w = c/f$, where c is the speed of light in vacuum. The peak boresight gain G in dBi is calculated as:

$$G(\text{dBi}) = 10 \log_{10} \left(\eta \left(\frac{\pi D}{\lambda_w} \right)^2 \right) \quad (12)$$

The aperture efficiency η collects the principal feed, illumination, blockage, and surface-error effects:

$$\eta = \eta_{sub} \cdot \eta_{spill} \cdot \eta_{taper} \cdot \eta_{surface} \quad (13)$$

where the factors represent blockage, spillover, illumination taper, and surface accuracy. A constant $\eta = 0.62$ is used when a frequency-specific datasheet gain is not supplied directly.

The Half-Power Beamwidth (θ_{HPBW}), which describes the angular width of the main beam and strongly affects adjacent-satellite interference susceptibility, is approximated by:

$$\theta_{\text{HPBW}} \approx 70^\circ \frac{\lambda_w}{D} \quad (14)$$

2.3 Free-Space Path Loss (FSPL) and Signal Spreading

Free-space path loss represents the reduction in power density caused by spherical spreading; it is not dissipative absorption by the vacuum.

Derived directly from the Friis Transmission Equation, the free-space path loss L_{FSPL} in decibel terms is expressed as:

$$L_{\text{FSPL}} [\text{dB}] = 20 \log_{10} \left(\frac{4\pi d}{\lambda_w} \right) = 92.45 + 20 \log_{10}(f_{\text{GHz}}) + 20 \log_{10}(d_{\text{km}}) \quad (15)$$

For the present GEO geometry, FSPL is approximately 206–207 dB.

2.4 Additional System Losses: Pointing Error and Polarization Mismatch

The clear-sky budget also includes fixed allowances for pointing, polarization mismatch, feeder loss, and miscellaneous implementation effects.

Antenna Pointing Error Loss (L_{point}): Wind, installation tolerance, structural deformation, and orbital motion can offset the antenna boresight. For a small pointing error θ_{error} , a Gaussian main-beam approximation gives:

$$L_{\text{point}} [\text{dB}] \approx 12 \left(\frac{\theta_{\text{error}}}{\theta_{\text{HPBW}}} \right)^2 \quad (16)$$

Polarization Mismatch Loss (L_{pol}): A fixed 0.3 dB allowance is used for residual feed alignment and polarization mismatch. Vertical linear polarization is selected for the representative P.838 calculation, but an operational link would use the polarization assigned to the leased transponder. The Type 243 datasheet specifies 30 dB on-axis cross-polarization discrimination [4]; rain depolarization and imperfect alignment can reduce this isolation.

2.5 RF Link Power Budget and Receiver Figure of Merit

For each link segment, EIRP combines transmitter output power, antenna gain, and feeder loss:

$$\text{EIRP} [\text{dBW}] = P_{\text{tx}} [\text{dBW}] + G_{\text{tx}} [\text{dBi}] - L_{\text{tx,feeder}} [\text{dB}] \quad (17)$$

Receiver performance is represented by G/T :

$$G/T [\text{dB/K}] = G_{\text{rx}} [\text{dBi}] - L_{\text{rx,feeder}} [\text{dB}] - 10 \log_{10} (T_{\text{sys}}) \quad (18)$$

where T_{sys} is the system noise temperature referred to the LNA input:

$$T_{\text{sys}} = T_{\text{ant}} + T_{\text{feeder}} + (F_{\text{LNA}} - 1)T_0 \quad (19)$$

Here T_{ant} includes sky and spillover contributions, T_{feeder} is the feeder contribution referred to the same plane, F_{LNA} is the LNA noise factor, and $T_0 = 290$ K. The static GS2 value of 150 K is a combined terminal assumption; it is not derived from a specified LNB noise figure.

The carrier-to-noise-density ratio at the receiver is:

$$(C/N_0) [\text{dBHz}] = \text{EIRP} [\text{dBW}] + G/T [\text{dB/K}] - L_{\text{total}} [\text{dB}] - 10 \log_{10}(k) \quad (20)$$

where L_{total} encompasses FSPL, pointing misalignments, polarization losses, and atmospheric clear-sky absorption. The term $-10 \log_{10}(k) \approx 228.6$ dB represents Boltzmann's constant in decibels.

For a fixed information bit rate R_b :

$$(E_b/N_0) [\text{dB}] = (C/N_0) [\text{dBHz}] - 10 \log_{10}(R_b) \quad (21)$$

The fixed-rate margin is the difference between the available and required E_b/N_0 :

$$M [\text{dB}] = (E_b/N_0) [\text{dB}] - (E_b/N_0)_{\text{req}} [\text{dB}] \quad (22)$$

2.6 Uplink-Downlink Coupling in Bent-Pipe Transponders

A transparent transponder does not regenerate the baseband signal. Uplink noise and interference therefore remain part of the retransmitted waveform.

The end-to-end $(C/N_0)_{\text{total}}$ is obtained by inverse-linear addition of the uplink and downlink terms:

$$\left(\frac{C}{N_0}\right)_{\text{total}}^{-1} = \left(\frac{C}{N_0}\right)_{\text{uplink}}^{-1} + \left(\frac{C}{N_0}\right)_{\text{downlink}}^{-1} \quad (23)$$

The combined C/N_0 is lower than either individual contribution. When one segment is much weaker than the other, the end-to-end result approaches the weaker segment and that path becomes the effective bottleneck.

3 Advanced Analysis Models

This chapter defines the optional impairment and diagnostic models added to the clear-sky calculation. Each model can be enabled independently, which also permits the ablation comparison reported in Chapter 5.

3.1 Adjacent-Satellite Interference (ASI) and IMD

Finite earth-station beamwidth permits energy from adjacent orbital slots to enter the receive chain and allows an uplink antenna to illuminate neighboring spacecraft. The resulting adjacent-satellite interference (ASI) depends on orbital spacing, antenna pattern, polarization, carrier loading, and relative EIRP. A multicarrier TWTA also generates intermodulation distortion (IMD) when operated close to saturation.

ASI and IMD are included as additional C/I terms so that the clear-sky result is not treated as purely thermal-noise limited.

The off-axis discrimination angle θ_{off} between the target satellite and the adjacent interferer is calculated using their respective ECEF position vectors. The off-axis gain $G_{\text{off}}(\theta_{\text{off}})$ in the direction of the interferer is estimated using a simplified ITU-R-style off-axis antenna envelope approximation:

$$G_{\text{off}}(\theta_{\text{off}}) = \min(G_{\text{max}} - 3.0, 32.0 - 25 \log_{10}(\theta_{\text{off}})) \quad (24)$$

This envelope is used for sensitivity analysis, not regulatory coordination. The uplink, downlink, and IMD C/I terms are combined with the thermal C/N terms by inverse-linear addition:

$$\left(\frac{C}{N+I}\right)_{\text{total}}^{-1} = \left(\frac{C}{N}\right)_{\text{uplink}}^{-1} + \left(\frac{C}{N}\right)_{\text{downlink}}^{-1} + \left(\frac{C}{I_{\text{ASI,up}}}\right)^{-1} + \left(\frac{C}{I_{\text{ASI,down}}}\right)^{-1} + \left(\frac{C}{I_{\text{IMD}}}\right)^{-1} \quad (25)$$

In this project, the ASI/IMD analysis is used as a parametric impairment estimate rather than a regulatory coordination calculation. The adjacent satellites are modeled at 11.0°E and 15.0°E, corresponding to a nominal 2.0° orbital separation from the target 13.0°E slot. The adjacent-carrier activity factor is set to −3.0 dB, the antenna off-axis rejection follows the simplified envelope above, and the representative transponder IMD term is modeled as $C/I_{\text{IMD}} = 30$ dB. A formal coordination study would require operator-specific carrier plans, measured antenna radiation patterns, polarization reuse information, and transponder loading data.

For the selected assumptions, ASI and IMD reduce the fixed-rate margin from 4.15 dB to 3.69 dB. The ablation study separates this total into approximately 0.44 dB from ASI and 0.01 dB from the representative IMD term. These values are scenario-specific and should not be used as coordination limits.

3.2 Hardware Impairments: Non-Linearity and Phase Noise

The transponder is not an ideal linear amplifier. TWTA compression, AM/PM conversion, and modem imperfections are represented only through simplified engineering terms:

- **Output Back-Off (OBO):** A deliberate reduction of the operating point below saturation to suppress non-linear distortion:

$$\text{EIRP}_{\text{active}} [\text{dBW}] = \text{EIRP}_{\text{sat}} [\text{dBW}] - \text{OBO} [\text{dB}] \quad (26)$$

- **Implementation Margin (M_{impl}):** In this study, modem implementation effects are not included as a static RF path loss. Instead, a 1.0 dB DVB-S2 implementation margin is applied only during MODCOD threshold selection:

$$(E_s/N_0)_{\text{usable}} = (E_s/N_0)_{\text{available}} - M_{\text{impl}} \quad (27)$$

An OBO value of 2.5 dB is discussed as a representative operating concept [7], but it is not subtracted separately from the published footprint EIRP: the 46 dBW downlink input is treated as the active contour value. Non-linear degradation in the numerical budget is represented by the assumed $C/I_{\text{IMD}} = 30$ dB term. A 1.0 dB modem implementation margin is applied only during DVB-S2 MODCOD selection, and the static RF implementation-loss field is set to 0 dB to avoid double counting.

3.3 Digital Baseband Performance Metrics

For coherent QPSK over AWGN, the uncoded bit-error probability is:

$$P_{b,\text{uncoded}} = \frac{1}{2} \text{erfc} \left(\sqrt{\frac{E_b}{N_0}} \right) \quad (28)$$

The report's generic digital-metrics layer applies a 3 dB equivalent coding-gain assumption for sensitivity calculations. The separate ACM calculation does not use this shortcut; it compares the available E_s/N_0 directly with the selected DVB-S2 threshold table. Shannon capacity is reported as an upper bound:

$$C(\text{bps}) = B \log_2 (1 + \text{SNR}) \quad (29)$$

This coding framework is the foundation for the DVB-S2 MODCOD ladder used by the ACM selector in Chapter 4.

3.4 Dynamic System Noise Temperature

The static budget uses a fixed T_{sys} . The dynamic model instead varies the antenna-noise contribution with elevation and with absorptive atmospheric loss. A lossy rain path both attenuates the carrier and increases the apparent sky temperature seen by the receive antenna.

Separating these effects avoids treating rain solely as a carrier-power loss. Chapter 5 reports the carrier attenuation first and then adds the receiver-noise increase as a separate ablation step.

The dynamic, elevation- and fade-dependent system noise temperature T_{sys} is modeled at the LNA input as:

$$T_{\text{sys}}(\theta, A_{\text{rain}}) = T_{\text{receiver}} + T_{\text{spillover}} + T_{\text{sky}}(\theta) + T_{\text{rain}}(A_{\text{rain}}) \quad (30)$$

where $T_{\text{sky}}(\theta)$ is the clear-sky noise at elevation angle θ , and T_{rain} is the rain thermal emission noise calculated using the radiative transfer equation:

$$T_{\text{rain}} = T_{\text{atm}} \left(1 - 10^{-A_{\text{rain}}/10} \right) \quad (31)$$

where $T_{\text{atm}} = 275$ Kelvin represents the physical atmospheric temperature of the lossy rain medium.

For example, 10 dB of absorptive loss gives $T_{\text{rain}} \approx 247.5$ K for $T_{\text{atm}} = 275$ K. This is the rain-emission contribution only; the total T_{sys} also depends on the receiver, spillover, and clear-sky terms and is calculated from the full expression above.

3.5 Apparent GEO Orbit Motion

The apparent spacecraft position is allowed to vary within a simplified station-keeping box:

$$\Delta\lambda(t) = A_{\text{ew}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ew}}\right) \quad (32)$$

$$\Delta\phi(t) = A_{\text{ns}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{ns}}\right) \quad (33)$$

$$\Delta r(t) = A_{\text{r}} \sin\left(\frac{2\pi t}{T_{\text{sidereal}}} + \phi_{\text{r}}\right) \quad (34)$$

These offsets update the slant-range vector $\vec{\rho}(t)$. With the modeled amplitudes (0.06° east–west, 0.04° north–south, and 40 km radial), the 48-hour margin variation is about 0.02 dB peak-to-peak. This result indicates that station-keeping motion is negligible in the present power budget; it does not establish a tracking requirement, which would depend on the measured antenna pattern, pointing tolerance, and operator practice.

4 Atmospheric Propagation and ACM

This chapter describes the atmospheric attenuation, illustrative outage, and DVB-S2 ACM calculations used after the clear-sky budget.

Model scope: the software is version-locked to the recommendation editions cited in this report, including P.618-13. Some recommendations have newer revisions, but changing versions would require a separate code and regression review. The present results use representative meteorological inputs rather than site-specific ITU digital-map extraction and should not be treated as an operational availability guarantee.

4.1 ITU-R Slant-Path Attenuation Suite

4.1.1 Slant-Path Loss Mechanisms

The Earth-space path crosses the troposphere, where gases, cloud liquid water, refractive-index fluctuations, and precipitation affect the received carrier. At Ku-band, rain is the largest time-varying attenuation term in the selected 99.9% design case.

4.1.2 Model Scope and Inputs

The ITU-R procedures are used to calculate excess attenuation beyond the fixed clear-sky allowances. Uplink and downlink attenuation are evaluated separately because their frequencies, elevations, and terminal apertures differ.

4.1.3 Mathematical Formulation

The total atmospheric attenuation $A_{\text{total}}(p)$ in dB exceeded for an annual time percentage p ($0.001\% \leq p \leq 5\%$) is calculated by combining gaseous absorption (A_{gaseous}), cloud attenuation (A_{cloud}), tropospheric scintillation ($A_{\text{scintillation}}$), and rain attenuation (A_{rain}) according to *ITU-R P.618-13* [8]:

$$A_{\text{total}}(p) = A_{\text{gaseous}} + \sqrt{(A_{\text{rain}}(p) + A_{\text{cloud}})^2 + A_{\text{scintillation}}(p)^2} \quad (35)$$

The point rainfall rate $R_{0.01}$ is a representative P.837-based input [9]. The same $R_{0.01} = 42$ mm/h value is used at both terminals; a site-specific study would use separate mapped values for Rome and Ankara. Rain height follows P.839-4 [?], while the frequency- and polarization-dependent k and α coefficients follow P.838-3 [10]. The specific attenuation is:

$$\gamma_R = k R_{0.01}^\alpha \quad (36)$$

The effective path length through the rain cell accounts for spatial localisation:

$$L_{\text{eff}} = L_s \cdot r_{0.01} \cdot v_{0.01} \quad (37)$$

where $r_{0.01}$ is the horizontal path reduction factor and $v_{0.01}$ is the vertical adjustment factor. The attenuation exceeded for 0.01% of the year is $A_{0.01} = \gamma_R L_{\text{eff}}$. For other time percentages p :

$$A_p = A_{0.01} \left(\frac{p}{0.01} \right)^{-[0.655 + 0.033 \ln p - 0.045 \ln A_{0.01} - \beta(1-p) \sin \theta]} \quad (38)$$

Gaseous attenuation uses dry air and water vapor specific zenith attenuations scaled by the slant-path elevation angle (*ITU-R P.676-12* [11]):

$$A_{\text{gaseous}} = \frac{\gamma_o h_o + \gamma_w h_w}{\sin \theta} \quad (39)$$

Cloud liquid water attenuation uses columnar liquid water M and specific attenuation K_1 (*ITU-R P.840-8* [12]):

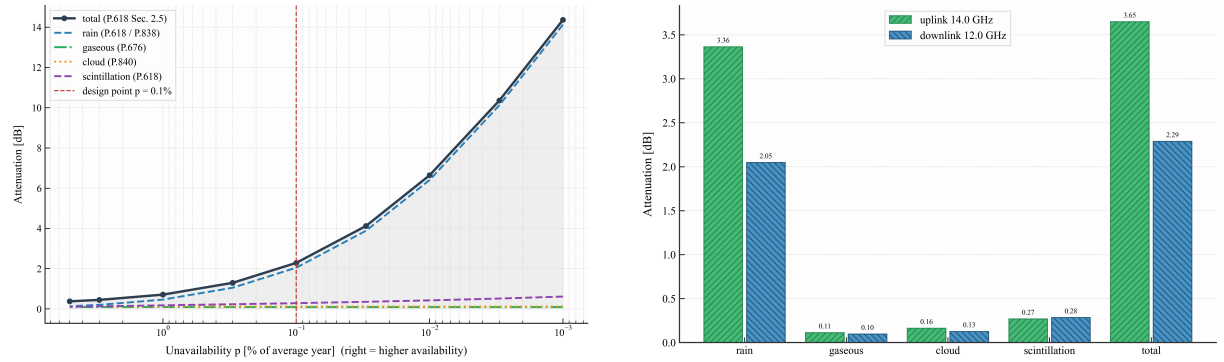
$$A_{\text{cloud}} = \frac{M \cdot K_1}{\sin \theta} \quad (40)$$

Scintillation uses the P.618-13 procedure with refractivity-related inputs consistent with P.453-14 [8, ?]:

$$A_{\text{scintillation}}(p) = a(p)\sigma \quad (41)$$

4.1.4 Operational Impact and Results

At 99.9% target availability, the model gives 3.65 dB total uplink attenuation at 14 GHz and 2.29 dB total downlink attenuation at 12 GHz. With the selected interference and dynamic-noise assumptions, the fixed-rate end-to-end margin becomes -0.02 dB.



(a) Slant-path attenuation vs. annual link availability.

(b) Attenuation breakdown at 99.9% availability.

Figure 6: ITU-R P.618 slant-path propagation impairments and specific contribution analysis for the Rome-Ankara link.

4.2 Monte-Carlo Rain Outage Simulation

4.2.1 Weather-State Model

The ITU-R design point gives attenuation exceeded for a specified percentage of an average year; it does not provide a chronological weather sequence. A separate Monte Carlo model is therefore used to generate an illustrative distribution of hourly margins.

4.2.2 Availability Estimation Rationale

This stochastic model is a sensitivity exercise based on assumed state probabilities and fade ranges. It is not fitted to measured precipitation records.

4.2.3 Mathematical Formulation

The simulation executes $N = 8760$ independent hourly trials. At each sample t_i , a weather state S_i is drawn from a probability mass function:

- S_{clear} : A small residual excess attenuation uniformly sampled between 0 and 0.15 dB.
- S_{light} **and** S_{moderate} : Small to medium fade depths drawn from uniform attenuation ranges.
- S_{heavy} : Rare high-fade states that represent severe Ku-band rain attenuation.

The discrete state variable is therefore:

$$S_i \in \{S_{\text{clear}}, S_{\text{light}}, S_{\text{moderate}}, S_{\text{heavy}}\}. \quad (42)$$

For each generated hourly weather state, the active link margin is recalculated as:

$$M_i = \left(\frac{E_b}{N_0}\right)_i - \left(\frac{E_b}{N_0}\right)_{\text{req}}. \quad (43)$$

The outage indicator for each sample is:

$$\mathcal{O}_i = \begin{cases} 1, & M_i < 0, \\ 0, & M_i \geq 0. \end{cases} \quad (44)$$

The corresponding link availability is then estimated as:

$$A_{\text{link}} = 100 \left(1 - \frac{1}{N} \sum_{i=1}^N \mathcal{O}_i\right) \%. \quad (45)$$

4.2.4 Operational Impact and Results

The output is a distribution of link margin under the assumed states and random seed. Because successive samples are independent, it does not reproduce storm duration or temporal correlation.

Note: This Monte Carlo model is an illustrative stochastic availability simulation based on assumed weather-state probabilities (light rain: 2.0%, moderate rain: 0.6%, heavy rain: 0.1%), not a site-calibrated meteorological time-series model derived from measured annual precipitation data. Results should therefore be interpreted as *estimated availability under the assumed stochastic weather-state model*, not as a prediction of actual annual availability.

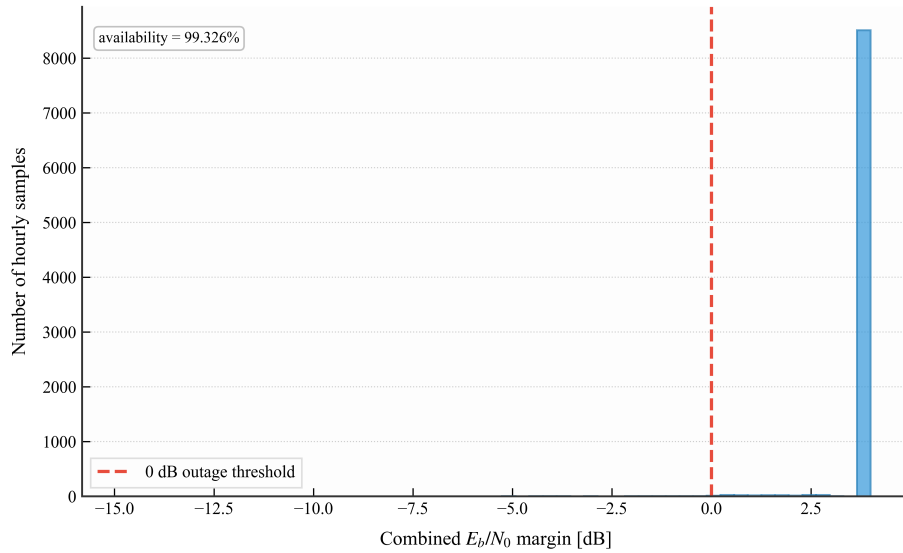


Figure 7: Probability density distribution of the dynamic link margin over 8,760 simulated hours of the meteorological year (Monte Carlo stochastic analysis).

4.3 DVB-S2 ACM (Adaptive Coding & Modulation)

4.3.1 Adaptive-Throughput Context

A fixed MODCOD must be selected to meet its target availability with adequate margin. When channel conditions improve, that fixed operating point may leave capacity unused.

4.3.2 Fixed-Rate and ACM Metrics

The fixed-rate link margin and the ACM throughput analysis are treated as separate performance views. The former evaluates closure against a prescribed (E_b/N_0) threshold, whereas the latter maps the available (E_s/N_0) to DVB-S2 MODCOD thresholds and net throughput for a specified carrier bandwidth and roll-off factor. The two metrics should not be interpreted as the same quantity.

ACM maps the measured E_s/N_0 to a modulation and code rate. Higher-efficiency modes are selected when margin is available; more robust modes are selected during fades.

4.3.3 Mathematical Formulation

The receiver continuously monitors $(E_s/N_0)_{\text{available}}$ and dynamically selects the highest spectral efficiency η (in bits/symbol) from the DVB-S2 [13] MODCOD ladder that satisfies:

$$(E_s/N_0)_{\text{available}} \geq (E_s/N_0)_{\text{MODCOD,req}} + M_{\text{impl}} \quad (46)$$

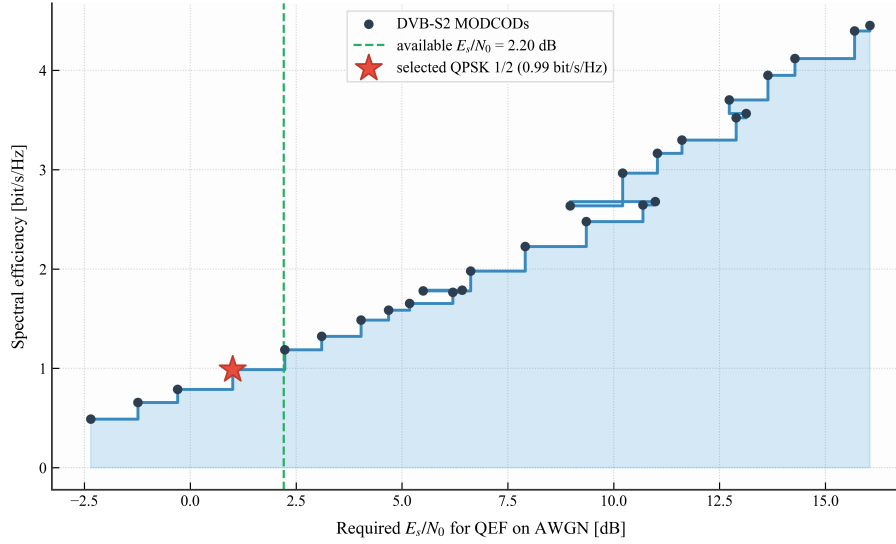


Figure 8: DVB-S2 MODCOD ladder: trade-off between spectral efficiency and required signal-to-noise ratio.

Using the usable symbol rate $R_s = B/(1 + \alpha_{\text{rolloff}})$, the active net throughput is:

$$R_{\text{net}} [\text{Mbps}] = \eta \cdot R_s \quad (47)$$

4.3.4 Operational Impact and Results

For a 36 MHz carrier with $\alpha = 0.20$, the symbol rate is 30 Msps. The implemented threshold table selects QPSK 3/4 for the clear-sky thermal and interference cases and QPSK 1/2 at the 99.9% atmospheric design point.

Table 3 lists the available and required E_s/N_0 values used by the selector.

Table 3: DVB-S2 ACM Consistency and Operating Point Verification

Case	Avail. E_s/N_0	Selected MODCOD	Req. E_s/N_0	Efficiency (η)	Net Throughput (ηR_s)
Clear-Sky (Thermal)	≈ 6.38 dB	QPSK 3/4	4.03 dB	1.487	≈ 44.61 Mbps
With ASI & IMD	≈ 5.92 dB	QPSK 3/4	4.03 dB	1.487	≈ 44.61 Mbps
99.9% Rain Fade	≈ 2.20 dB	QPSK 1/2	1.00 dB	0.989	≈ 29.67 Mbps

Note: All points assume 1.0 dB implementation margin. Throughput is computed using $R_s = 30$ Msps.

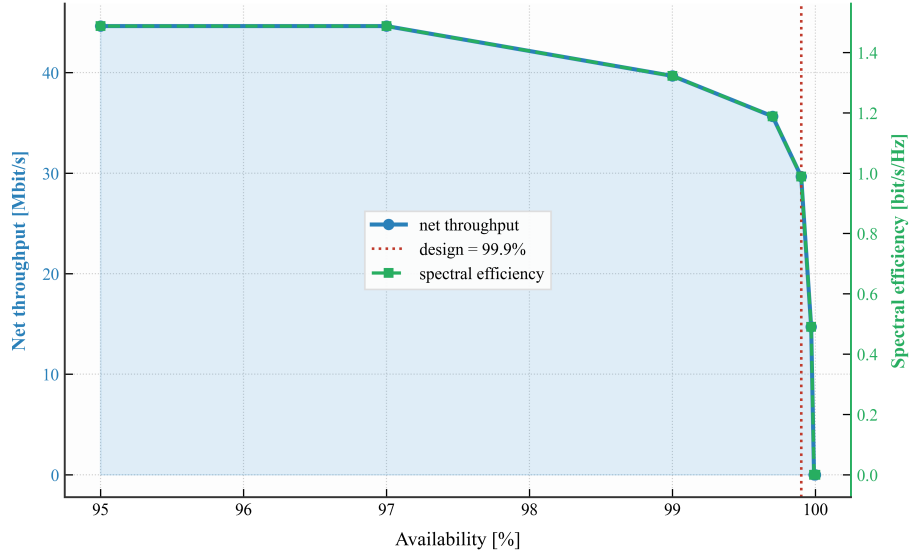


Figure 9: DVB-S2 ACM dynamic throughput and MODCOD selection over the link availability spectrum.

4.4 Uplink Power Control (ULPC)

Uplink power control is discussed as a mitigation option but is not implemented as a closed-loop simulation. An ideal first-order controller would increase earth-station transmit power by the estimated uplink fade:

$$P_{\text{tx,dynamic}} [\text{dBW}] = P_{\text{tx,nominal}} [\text{dBW}] + A_{\text{rain,up}} [\text{dB}] \quad (48)$$

subject to $P_{\text{tx,dynamic}} \leq P_{\text{tx,max}}$. Available HPA headroom, EIRP limits, ASI constraints, transponder loading, and control-loop delay limit the recoverable fade. No ULPC gain is credited in the reported margins.

5 Numerical Results and Discussion

This chapter reports the computed operating points, validation checks, sensitivity studies, and module-ablation results.

5.1 Static Clear-Sky Link Budget Results

The clear-sky calculation uses the Rome–HOTBIRD 13G–Ankara geometry defined in Chapter 1.

Table 4 distinguishes published hardware or operator values from model assumptions and explains how each value enters the calculation.

Table 4: Hardware and operator data traceability.

Item	Published value	Model value	Treatment
Andrew Type 243 antenna [4]	2.4 m aperture; 48.9 dBi at 14.3 GHz; 13.75–14.50 GHz Tx	2.4 m, $\eta = 0.62$	Computed gain is 48.86 dBi at 14.0 GHz, consistent with the datasheet.
Norsat ATOMBKU020 [5]	$P_{\text{1dB}} \geq 43$ dBm; 14.0–14.5 GHz standard band	20 W nominal output	The 20 W class is used; no additional amplifier gain is invented.
HOTBIRD 13F/13G widebeam [1]	13°E; official 40–53 dBW contour map	46 dBW at Ankara	Conservative map reading; not a transponder-specific guaranteed value.
Satellite receive G/T	Not published for the selected HOTBIRD transponder	3 dB/K	Explicit engineering assumption.
Triax TD88 [6]	95 cm $\times$ 85 cm; 38.8 dBi at 11.7 GHz	0.9 m equivalent circular aperture	Computed gain is approximately 39.0 dBi at 12 GHz.
GS2 receiver noise	LNB-specific combined T_{sys} unavailable	150 K static baseline	Explicit assumption; dynamic model is evaluated separately.
Fixed-rate threshold	Scenario design requirement	Required $E_b/N_0 = 7$ dB	Used only for the 10 Mbps fixed-rate closure test.

Table 5 gives the intermediate values from the software implementation of the Pratt-style calculation [7].

Table 5: Static Clear-Sky Link Budget Results (Rome - Ankara).

Link Segment	Parameter	Uplink	Downlink	Unit
Geometry	Coordinates	41.90°N, 12.50°E	39.93°N, 32.86°E	–
	Satellite Longitude	13.00°E	13.00°E	–
	Slant Range (d)	37,658.79	37,823.12	km
	Elevation Angle (θ)	41.60	39.44	deg
	Azimuth Angle (Az)	179.25	209.37	deg
	Central Angle	41.91	43.85	deg
Transmitter	Frequency (f)	14.00	12.00	GHz
	Antenna Diameter	2.40	–	m
	Antenna Gain	48.86	–	dBi
	Transmitter Power	13.01 (20.0 W)	–	dBW
	EIRP	60.87	46.00 (Override)	dBW
Path Losses	Free Space Path Loss (FSPL)	206.89	205.59	dB
	Pointing Loss	0.50	0.50	dB
	Polarization Loss	0.30	0.30	dB
	Clear-Sky Atmospheric Loss	0.50	0.50	dB
	Misc Loss	0.20	0.20	dB
	Total Link Loss (L_{total})	208.39	207.09	dB
Receiver	Antenna Diameter	–	0.90	m
	Antenna Gain	–	39.00	dBi
	Noise Temperature (T_{sys})	–	150.00	K
	Quality Factor (G/T)	3.00 (Override)	16.74	dB/K
Performance	Carrier-to-Noise Density (C/N_0)	84.08	84.25	dB-Hz
	Carrier-to-Noise Ratio (C/N)	8.51	8.69	dB
	Energy-per-Bit to Noise (E_b/N_0)	14.08	14.25	dB
	Required E_b/N_0	7.00	7.00	dB
	Link Margin	7.08	7.25	dB

The end-to-end combined carrier-to-noise density ratio is computed using inverse-linear addition:

$$\left(\frac{C}{N_0}\right)_{\text{total,linear}} = \left(10^{-84.08/10} + 10^{-84.25/10}\right)^{-1} \quad (49)$$

$$\left(\frac{C}{N_0}\right)_{\text{total,dBHz}} = 10 \log_{10} \left(\left(\frac{C}{N_0}\right)_{\text{total,linear}} \right) = 81.15 \text{ dB-Hz} \quad (50)$$

This gives $E_b/N_0 = 11.15$ dB and a 10 Mbps fixed-rate margin of **4.15 dB** against the 7 dB requirement.

Table 6 checks the implementation against an independent substitution of the same link-budget equations.

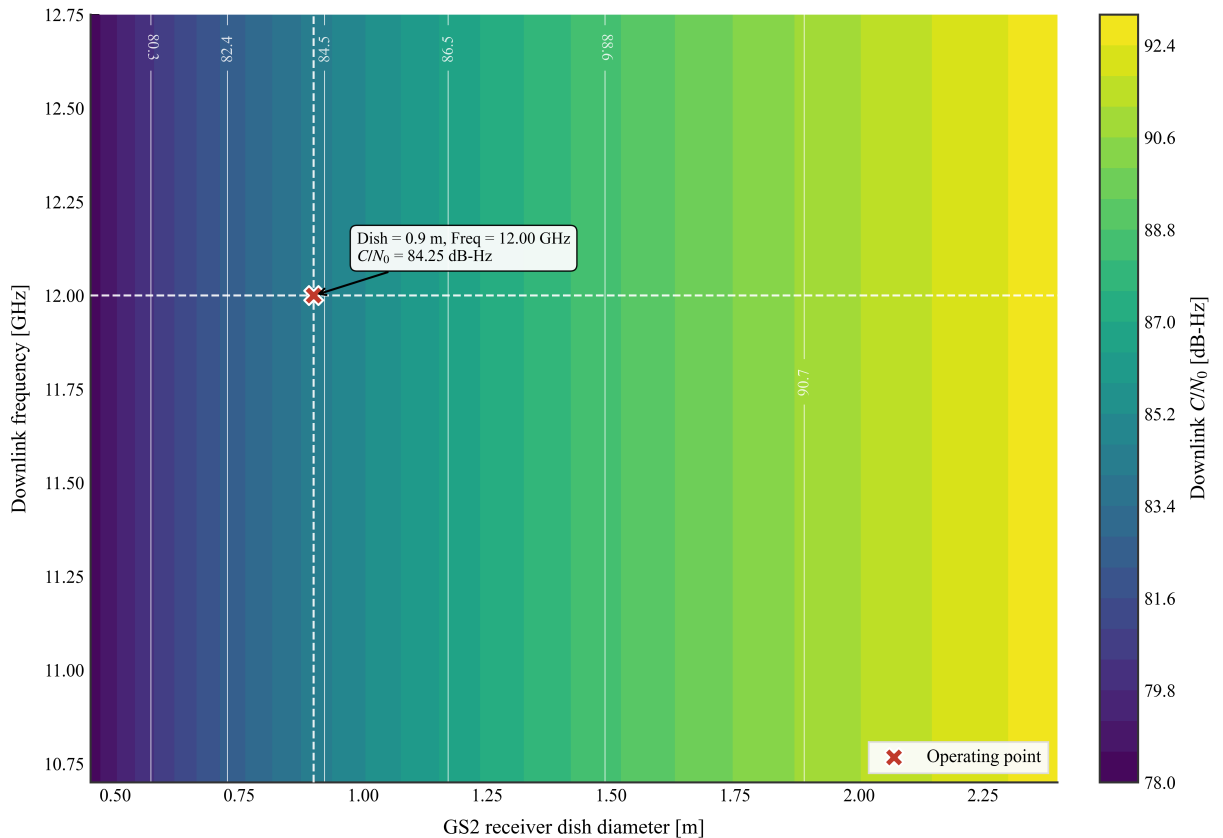
Table 6: Validation: Manual Pratt Calculation vs. Python Output.

Quantity	Hand calculation	Python output	Difference
Uplink FSPL	206.89 dB	206.89 dB	0.00 dB
Downlink FSPL	205.59 dB	205.59 dB	0.00 dB
Uplink EIRP	60.87 dBW	60.87 dBW	0.00 dB
Downlink G/T	16.74 dB/K	16.74 dB/K	0.00 dB
Uplink C/N_0	84.08 dB-Hz	84.08 dB-Hz	0.00 dB
Downlink C/N_0	84.25 dB-Hz	84.25 dB-Hz	0.00 dB
Uplink C/N	8.51 dB	8.51 dB	0.00 dB
Downlink C/N	8.69 dB	8.69 dB	0.00 dB
Combined C/N_0	81.15 dB-Hz	81.15 dB-Hz	0.00 dB
Combined E_b/N_0	11.15 dB	11.15 dB	0.00 dB
Combined margin	4.15 dB	4.15 dB	0.00 dB

5.1.1 Parametric Sensitivity and 2D Contour Map Analysis

Five contour plots show the principal static sensitivities and link-closure boundaries.

5.1.2 Downlink (C/N_0) vs. GS2 Dish Diameter and Frequency


 Figure 10: Downlink (C/N_0) vs. GS2 dish diameter and frequency.

- **Purpose:** Evaluate downlink C/N_0 over receive-dish diameters of 0.5–2.4 m and frequencies of 10.7–12.75 GHz.
- **Result:** C/N_0 ranges from approximately 81.5 to 91.5 dB-Hz over the plotted domain. The nearly vertical contours show that aperture is the dominant variable.
- **Interpretation:** For fixed physical aperture and EIRP, receive gain and FSPL both contain an f^2 dependence, so their frequency terms largely cancel. Frequency becomes important when atmospheric loss, efficiency, or footprint EIRP also varies with band.
- **Design use:** The plot gives the minimum receive aperture needed for a specified C/N_0 under the fixed assumptions.

5.1.3 Downlink Margin vs. Satellite EIRP and GS2 T_{sys}

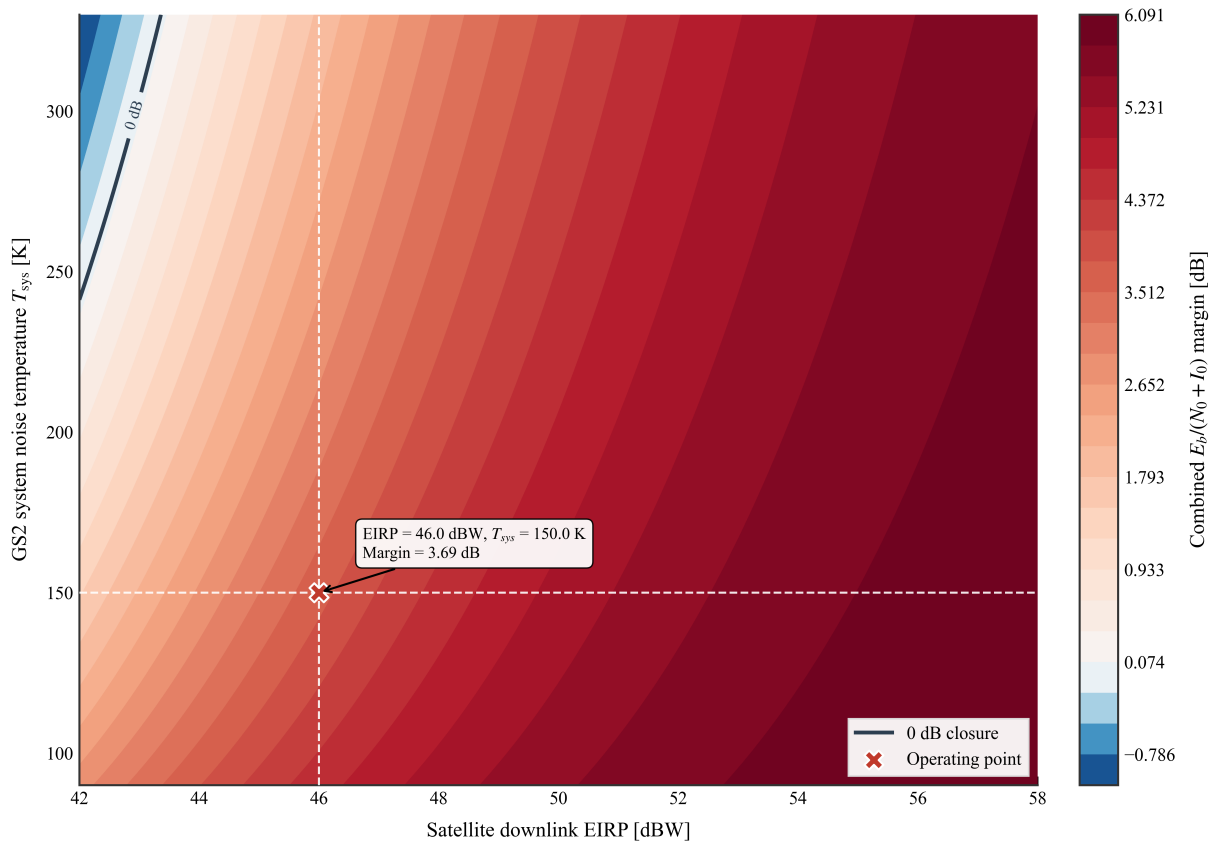


Figure 11: Combined effective margin vs. satellite EIRP and GS2 T_{sys} .

- **Purpose:** Map the 0 dB end-to-end margin boundary over 42–58 dBW downlink EIRP and 90–330 K receiver noise temperature, with ASI and IMD enabled.
- **Numerical result:** The 0 dB contour separates closed and open combinations. At $T_{\text{sys}} = 300$ K, closure requires approximately 45.5 dBW. The selected 46 dBW, 150 K point gives approximately 3.69 dB with ASI and IMD enabled.
- **Interpretation:** Margin increases one-for-one with EIRP in dB, while the receiver term varies as $-10 \log_{10} T_{\text{sys}}$.

- **Design use:** The map quantifies the trade between leased downlink EIRP and receive-terminal G/T .

5.1.4 Uplink C/N vs. HPA Power and GS1 Dish Diameter

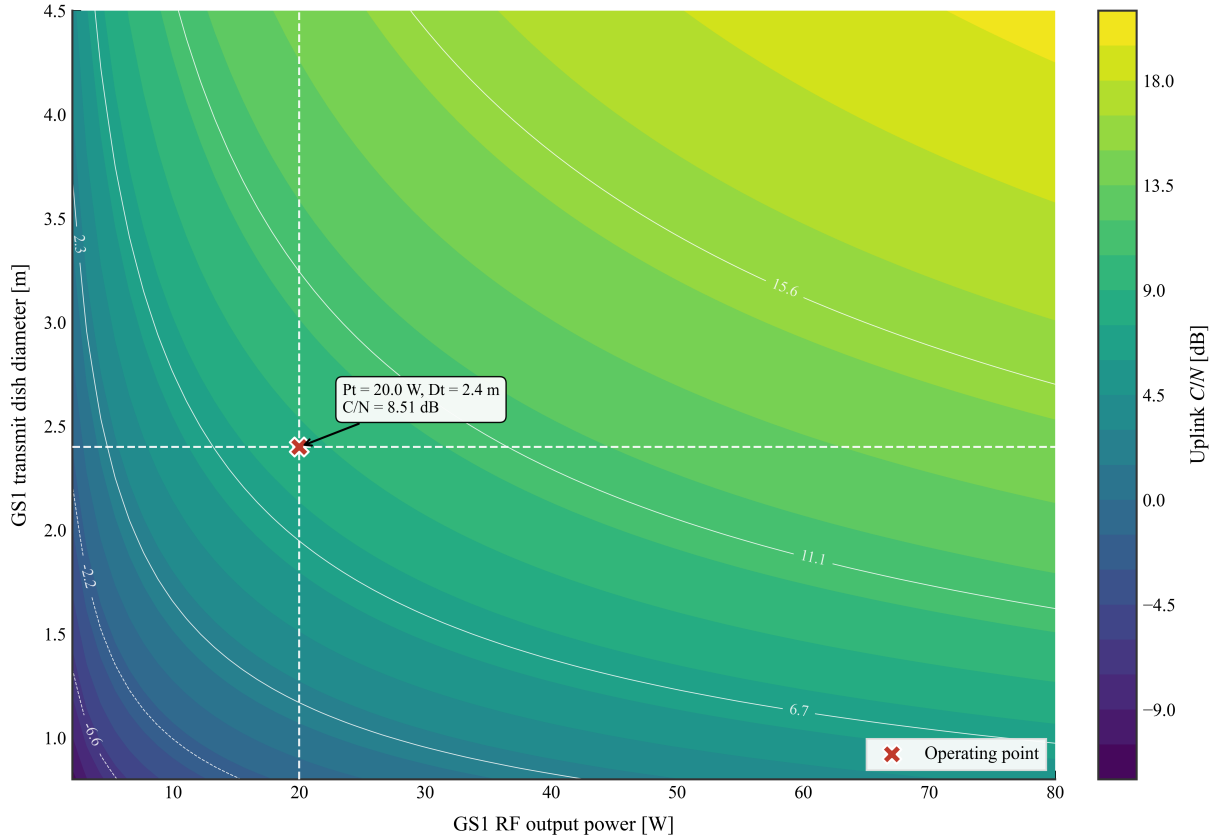


Figure 12: Uplink C/N vs HPA Power and GS1 Dish Diameter.

- **Purpose:** Compare 2–80 W HPA output with 0.8–4.5 m uplink aperture.
- **Result:** The 2.4 m, 20 W baseline gives $C/N \approx 8.51$ dB. A 12 dB target at the same aperture requires about 45 W. At 1.2 m, the additional 6 dB aperture penalty would require about 180 W, outside the plotted range.
- **Interpretation:** Doubling dish diameter adds approximately 6 dB of gain. For this case, aperture growth is more effective than increasing HPA output alone.
- **Design use:** The plot supports a cost and packaging trade between antenna size, amplifier rating, and off-axis discrimination.

5.1.5 Combined (E_b/N_0) vs. Information Bit Rate and Bandwidth

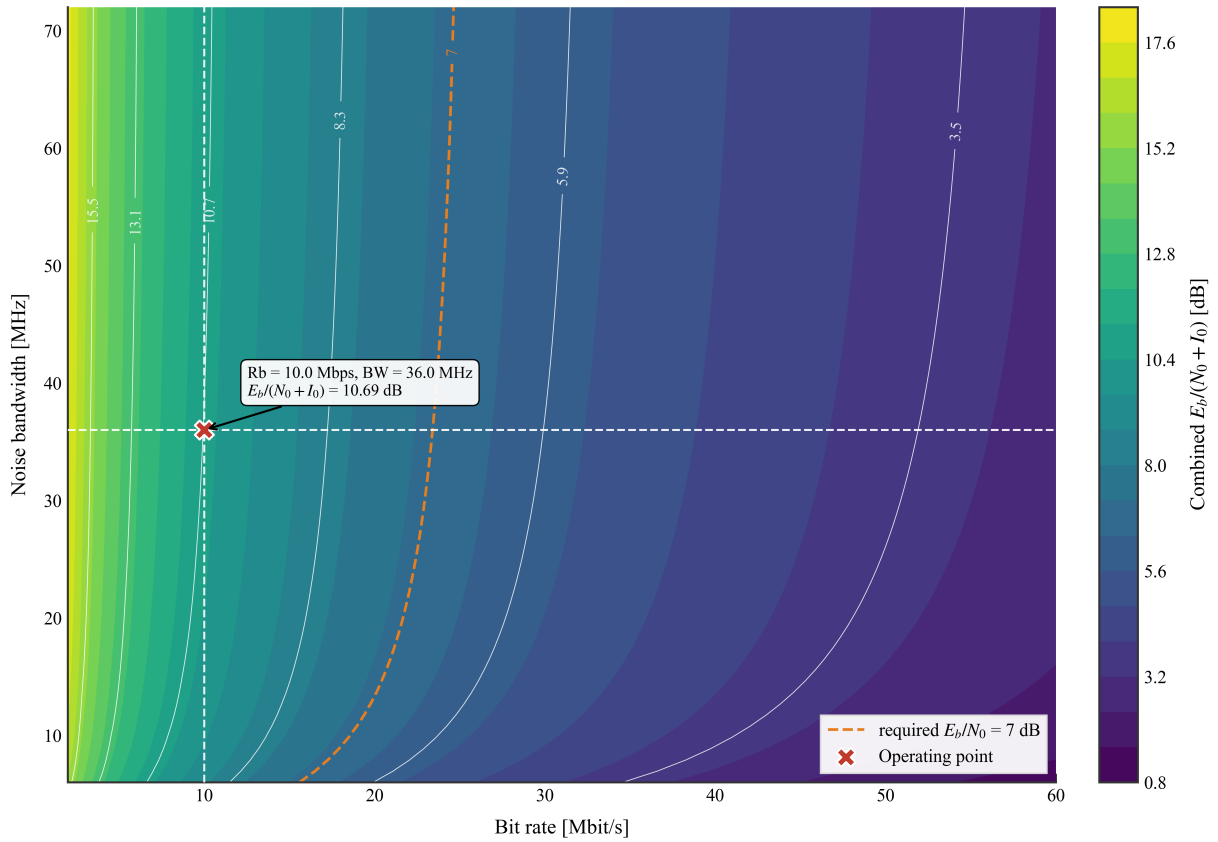


Figure 13: Effective end-to-end $E_b/(N_0 + I_0)$ vs. information bit rate and bandwidth.

- **Purpose:** Evaluate effective end-to-end E_b/N_0 over 2–60 Mbps and 6–72 MHz.
- **Result:** With $C/N_0 \approx 81.15$ dB-Hz, the noise-only E_b/N_0 falls from about 18.1 dB at 2 Mbps to 4.2 dB at 50 Mbps. At 10 Mbps it is 11.15 dB, or 10.69 dB with ASI and IMD.
- **Interpretation:** At fixed C/N_0 , E_b/N_0 depends on bit rate, not directly on allocated bandwidth. Bandwidth instead limits symbol rate, occupied spectrum, and feasible MODCODs.
- **Design use:** The plot identifies the fixed-rate closure boundary for the stated 7 dB requirement.

5.1.6 GS2 Elevation Angle vs. Latitude and Longitude Separation

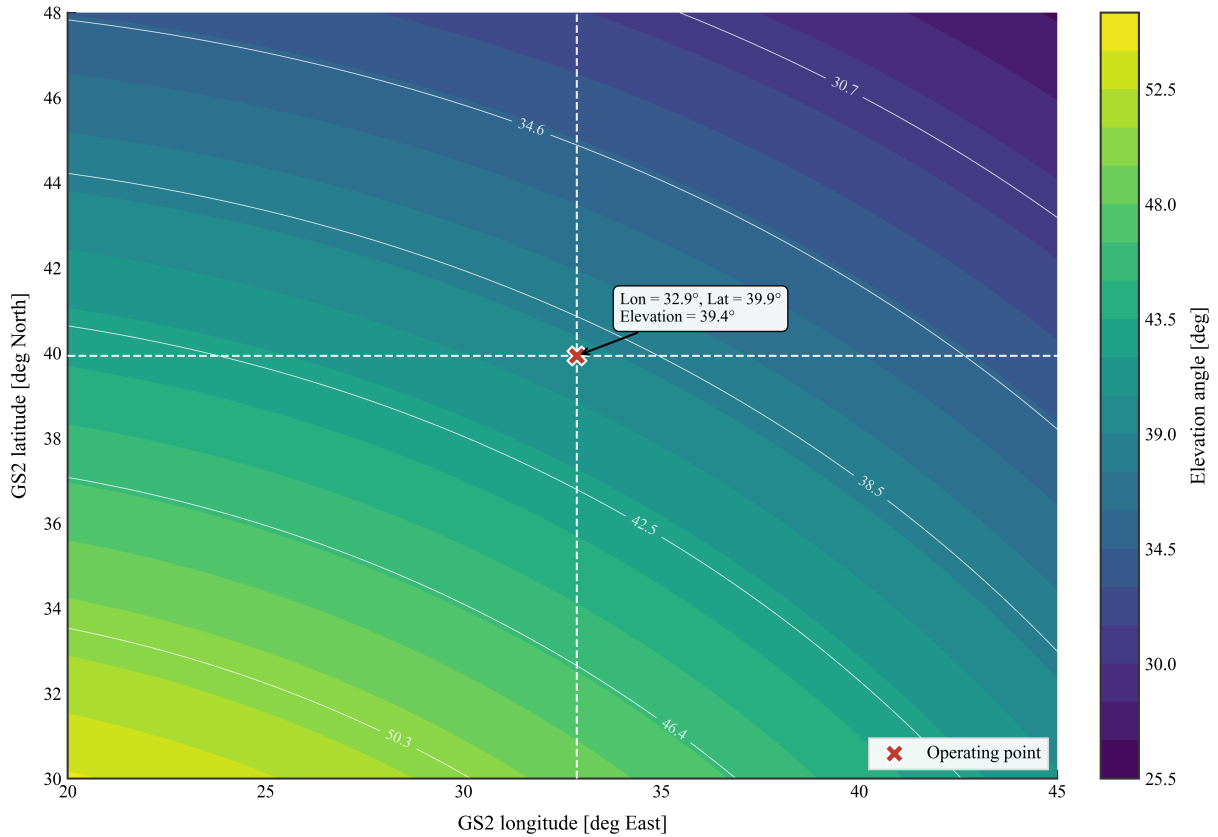


Figure 14: GS2 Elevation Angle vs Latitude and Longitude Separation.

- **Purpose:** Map GEO elevation angle over station latitude and longitude separation.
- **Result:** Elevation decreases as either latitude or longitude separation increases; 10-degree and 5-degree reference contours appear near the edge of the plotted domain.
- **Interpretation:** Lower elevation increases slant range and the atmospheric path length.
- **Design use:** The map indicates where additional atmospheric and pointing allowances become necessary.

5.2 Advanced Impairment and Dynamic Analysis

This section adds the optional interference, orbital, atmospheric, and adaptation models to the static reference case.

5.2.1 Interference (ASI and IMD)

- **Purpose:** Evaluate ASI from modeled neighbors at 11°E and 15°E and add the assumed transponder IMD term.
- **Numerical result:** ASI and IMD reduce the margin from 4.15 dB to **3.69 dB**. The ablation study attributes about 0.44 dB to ASI and 0.01 dB to the assumed IMD term.

- **Interpretation:** The combined 0.46 dB penalty is modest for the selected antenna sizes and assumed activity factor, but it is not a coordination result.
- **Design use:** An operational study would replace the generic sidelobe envelope and IMD assumption with measured antenna patterns, assigned polarizations, carrier powers, and transponder loading.

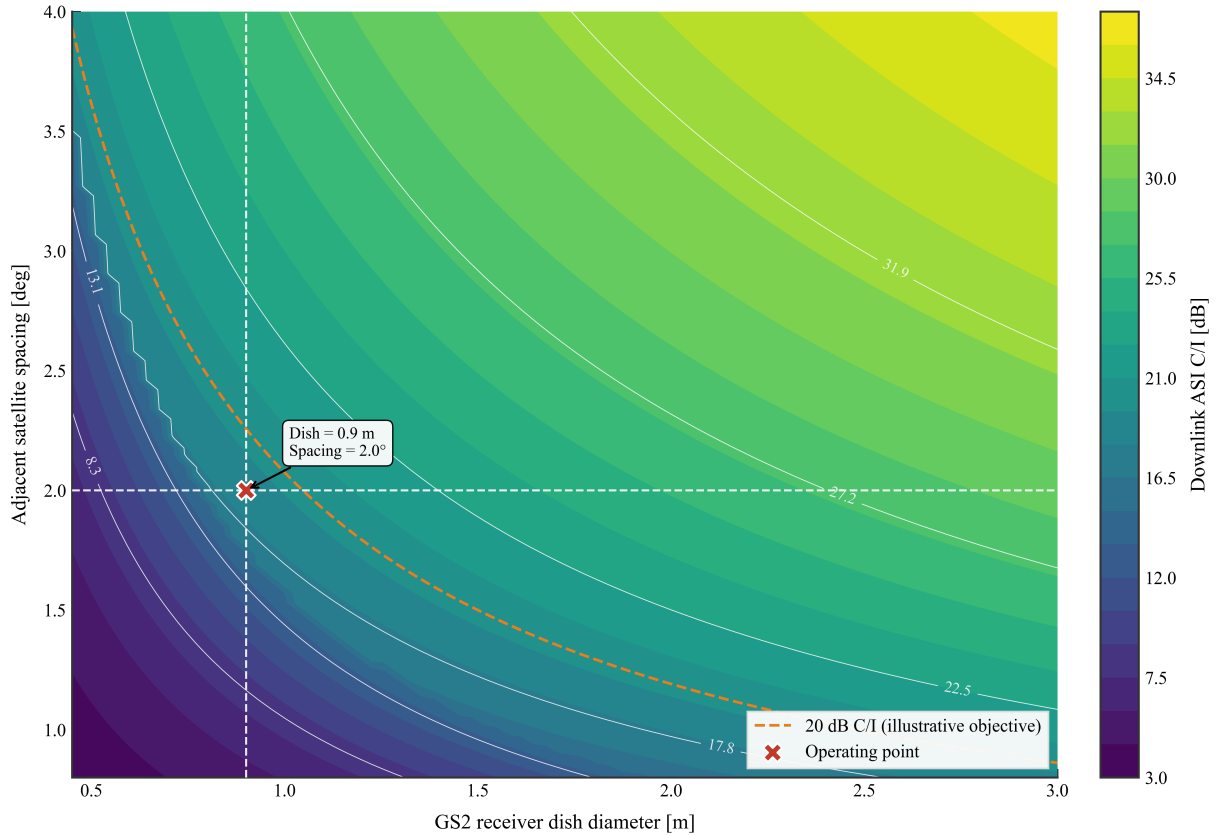


Figure 15: Downlink Adjacent Satellite Interference (ASI) C/I ratio as a function of orbital spacing and receiving antenna diameter.

5.2.2 Time-Varying Orbital Dynamics

- **Purpose:** Apply the simplified station-keeping motion model over 48 hours.
- **Result:** The combined margin varies by approximately 0.02 dB peak-to-peak.
- **Interpretation:** The modeled longitude, latitude, and radial excursions produce only small range and elevation changes for this GEO geometry.
- **Design use:** The approximately 0.02 dB variation is negligible for this budget. The result is a consistency check for the GEO motion model, not a validation for LEO or MEO orbit propagation.

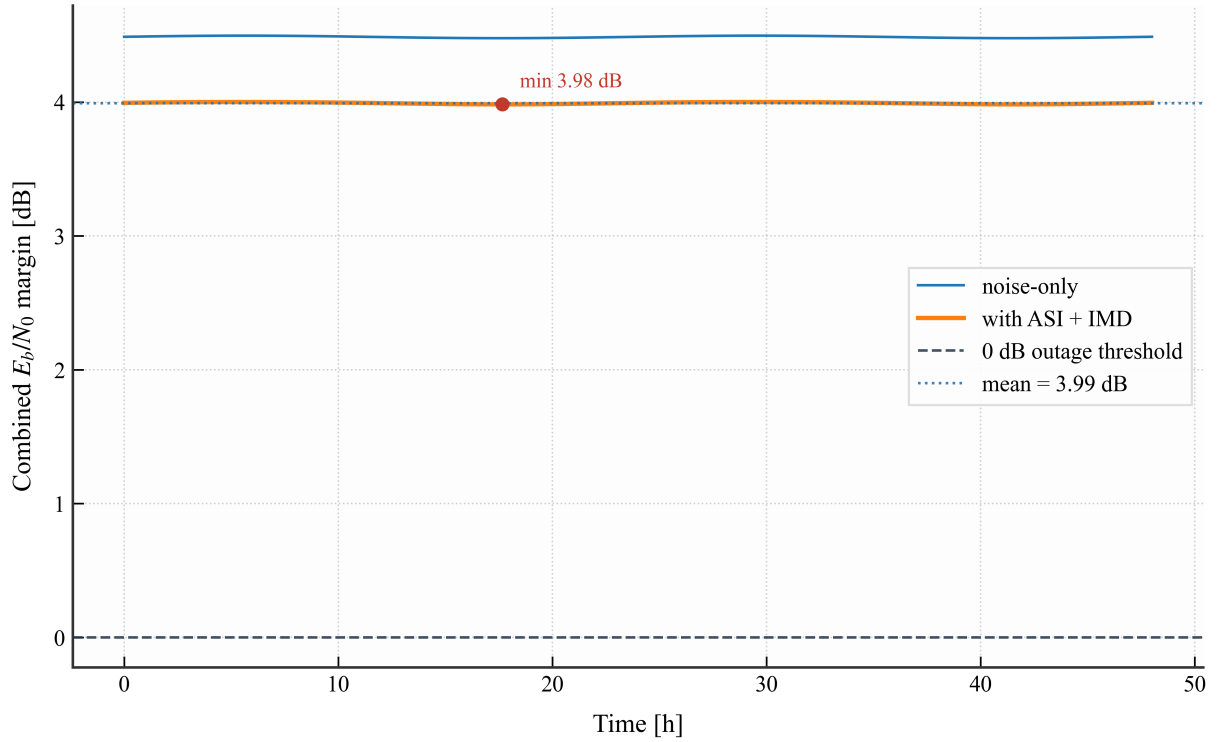


Figure 16: Time-varying combined link margin under the modeled apparent satellite motion.

5.2.3 ITU-R Rain Fade and Atmospheric Impairments

- **Purpose:** Calculate the excess attenuation at the 99.9% target-availability point ($p = 0.1\%$).
- **Result:**
 - **Uplink Attenuation (14.0 GHz):** Total combined atmospheric fade of 3.65 dB.
 - **Downlink Attenuation (12.0 GHz):** Total combined atmospheric fade of 2.29 dB.
- **Interpretation:** Rain contributes about 90% of the modeled excess attenuation; gas, cloud, and scintillation make up the remainder.
- **Design use:** The resulting -0.02 dB margin does not close the 10 Mbps fixed-rate case. Additional RF margin or a revised service requirement is needed for the stated target.

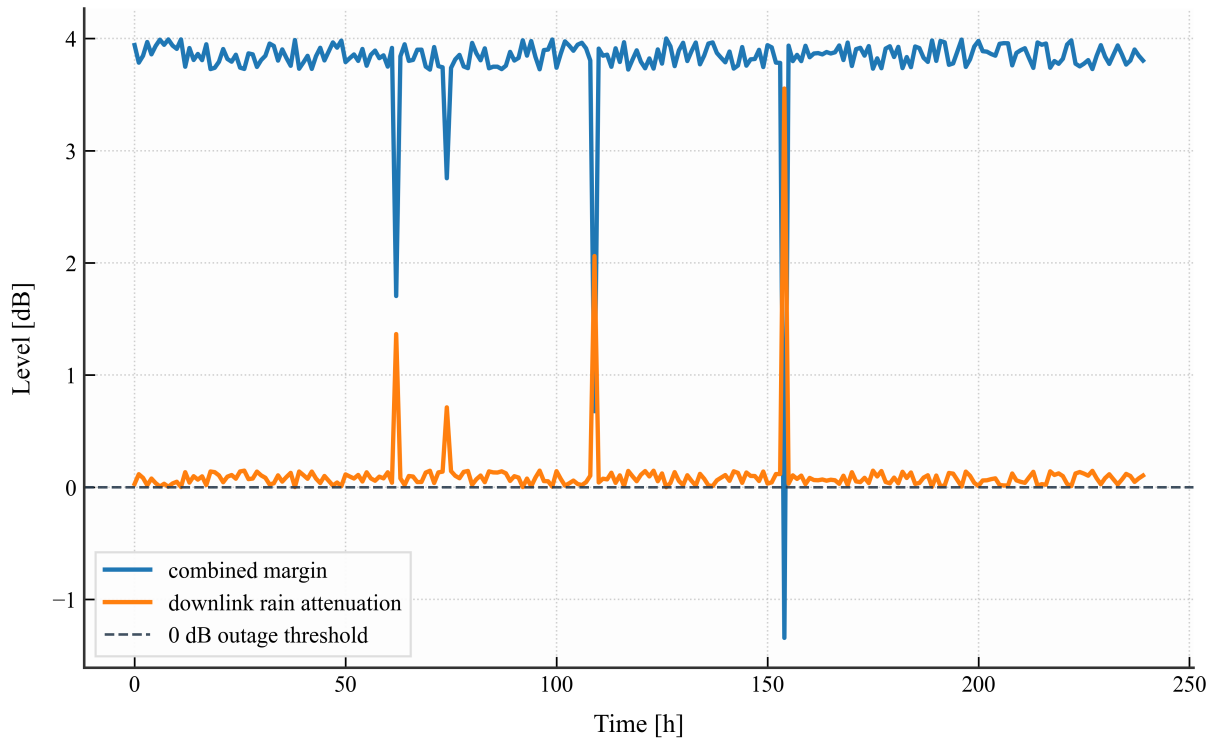


Figure 17: Illustrative link-margin response when the stochastic model applies heavy attenuation to both link directions.

5.2.4 DVB-S2 ACM Throughput Performance

- **Purpose:** Apply the DVB-S2 selector at the 99.9% atmospheric design point.
- **Numerical result:** At the 99.9% atmospheric design point, the selector chooses **QPSK 1/2**, with a tabulated threshold of $E_s/N_0 = 1.00$ dB and estimated net throughput of **29.67 Mbps**.
- **Interpretation:** QPSK 1/2 has a lower threshold than QPSK 3/4 and therefore remains selectable at reduced throughput.
- **Design use:** ACM preserves a variable-rate operating mode; it does not add physical link margin or prove fixed-rate availability.

5.3 Performance Comparison and Engineering Synthesis

The static and extended calculations use the same hardware and geometry inputs; only the enabled impairment and adaptation models change.

Design Implication: Fixed-Rate Margin vs. ACM Throughput

Fixed-rate metric: $E_b/N_0 = C/N_0 - 10 \log_{10}(R_b)$

ACM metric: $E_s/N_0 = C/N_0 - 10 \log_{10}(R_s)$

Symbol rate: $R_s = B/(1 + \alpha)$

Effective impaired metric: Thermal noise and interference are combined using inverse-linear summation.

Clarification: The fixed-rate 10 Mbps case and the DVB-S2 ACM throughput case are not evaluated with the same closure criterion. The fixed-rate case uses a conservative required $E_b/N_0 = 7$ dB design threshold. The ACM case converts the available end-to-end $C/(N + I)_0$ into E_s/N_0 at the DVB-S2 symbol rate and checks closure against MODCOD-specific DVB-S2 thresholds plus implementation margin. Therefore, the ACM throughput values are estimated variable-rate DVB-S2 carrier capacities under the selected MODCOD, not a direct continuation of the fixed 10 Mbps operating mode.

5.3.1 Module Ablation Study

The three-state comparison is complemented by a controlled ablation study in which one physical effect is introduced at a time. The values are generated by the same analysis functions and exported to `outputs/ablation_summary.csv`; therefore, the table is not a collection of independent hand-picked cases. The cumulative rows isolate the incremental margin change caused by each model, while the diagnostic rows report effects that need not be one-sided impairments.

Table 7: Ablation study separating the contribution of each advanced model.

Stage	Active Effects	10 Mbps Margin	Increment	GS2 T_{sys}	UL/DL Excess Loss	Margin Range / Variation	Result
A0	Thermal-noise-only static baseline	4.15 dB	reference	150 K	0/0 dB	single point	Fixed-rate closed
A1	A0 + adjacent-satellite interference	3.71 dB	−0.44 dB	150 K	0/0 dB	single point	ASI is the main clear-sky interference penalty
A2	A1 + transponder IMD	3.69 dB	−0.01 dB	150 K	0/0 dB	single point	IMD adds a small penalty for the assumed 30 dB C/I term
A3	A2 + gas, cloud, and scintillation; fixed T_{sys}	3.32 dB	−0.38 dB	150 K	0.43/0.41 dB	single point	Non-rain atmospheric loss remains moderate
A4	A3 + rain attenuation at 99.9% availability; fixed T_{sys}	0.88 dB	−2.43 dB	150 K	3.65/2.29 dB	single point	Carrier attenuation consumes most of the remaining margin
A5	A4 + dynamic rain-emission noise temperature	−0.02 dB	−0.91 dB	239 K	3.65/2.29 dB	single point	Fixed-rate 10 Mbps no longer closes
A6	A5 + DVB-S2 ACM response	−0.02 dB	0.00 dB	239 K	3.65/2.29 dB	single point	QPSK 1/2, approximately 29.67 Mbps; physical margin is unchanged
D0	ASI + IMD + dynamic clear-sky T_{sys}	3.99 dB	+0.30 dB vs. A2	127 K	0/0 dB	single point	The fixed 150 K baseline is conservative in clear sky
D1	D0 + 48-hour GEO station-keeping motion	nominal 3.99 dB	diagnostic	≈ 127 K	0/0 dB	3.98–4.00 dB; 0.017 dB peak-to-peak	Orbital motion is negligible for this GEO link

The ablation sequence shows that the advanced result is not driven by a single opaque “advanced-model penalty.” ASI accounts for approximately 0.44 dB, whereas the selected IMD assumption contributes only about 0.01 dB. At the 99.9% design point, rain-induced carrier attenuation is the dominant incremental loss, and the associated increase in downlink sky-noise temperature contributes a further 0.91 dB penalty. ACM does not restore the fixed-rate physical margin; it changes the operating threshold and throughput. The separate GEO-motion diagnostic produces only about 0.017 dB peak-to-peak margin variation.

Table 8 then condenses the detailed ablation sequence into the three principal operational states used throughout the report.

Table 8: Performance Comparison: Static Baseline vs. Advanced Impaired Conditions.

Metric	Clear-Sky (Noise Only)	With ASI & IMD	With 99.9% Rain Fade
Link Margin	4.15 dB	3.69 dB	−0.02 dB
Available E_b/N_0 at 10 Mbps	≈ 11.15 dB	≈ 10.69 dB	≈ 6.98 dB
Available E_s/N_0 for ACM	≈ 6.38 dB	≈ 5.92 dB	≈ 2.20 dB
Supported MODCOD	QPSK 3/4	QPSK 3/4	QPSK 1/2
Availability interpretation	Not evaluated	Clear-sky snapshot	99.9% target point
Fixed-rate 10 Mbps closure	Closed	Closed	Not closed
ACM Net Throughput	≈ 44.6 Mbps	≈ 44.6 Mbps	29.67 Mbps

Note: The throughput values in Table 8 represent the achievable DVB-S2 ACM net throughput for the allocated 36 MHz carrier, not the baseline 10 Mbps information-rate setting used in the fixed-rate calculations.

Interpretation of the Comparison:

1. **Interference:** The selected ASI and IMD assumptions impose a combined penalty of approximately 0.46 dB. A coordination-grade value would require measured patterns and operator carrier data.
2. **Atmospheric design point:** At 99.9% target availability, the fixed-rate margin becomes -0.02 dB. The baseline hardware therefore needs additional EIRP, receive aperture, power-control headroom, or a lower availability or rate requirement to close with the stated 7 dB threshold.
3. **ACM response:** QPSK 1/2 remains selectable at the atmospheric design point and gives 29.67 Mbps for the 36 MHz carrier. This is a separate variable-rate operating mode and does not change the negative 10 Mbps fixed-rate margin.

The clear-sky 10 Mbps link closes with positive margin. The selected interference assumptions cause a modest reduction, while the 99.9% atmospheric case consumes the remaining fixed-rate margin. ACM provides a lower-threshold variable-rate mode, but the baseline configuration still does not demonstrate 99.9% fixed-rate closure.

6 Conclusion

The report evaluated a 14/12 GHz transparent GEO link from Rome to Ankara through HOTBIRD 13G. The same Python model was used for the clear-sky budget, the optional impairment cases, and the ablation comparison.

In clear sky, the combined C/N_0 is 81.15 dB-Hz and the 10 Mbps fixed-rate margin is 4.15 dB. The selected ASI and IMD terms reduce the margin to 3.69 dB. At the 99.9% atmospheric design point, the margin is -0.02 dB; the stated baseline therefore does not close against the 7 dB fixed-rate threshold at that availability.

For the 36 MHz carrier and 0.20 roll-off factor, the ACM selector changes from QPSK 3/4 in clear sky to QPSK 1/2 at the atmospheric design point. The latter corresponds to an estimated 29.67 Mbps and remains a variable-rate mode, not evidence of 10 Mbps fixed-rate availability.

The largest modeled penalties come from rain attenuation and the associated increase in receive-system noise temperature. Candidate design changes are a larger GS2 aperture, higher contracted downlink EIRP, verified uplink power-control headroom, or a lower fixed-rate requirement. An operational design would also replace the shared representative rainfall rate with separate mapped or measured values for Rome and Ankara and would use transponder-specific EIRP, G/T, polarization, and loading data.

Intelsat 39 and other GEO spacecraft may be suitable for different coverage or connectivity objectives. For the sites and broadcast-oriented service considered here, HOTBIRD 13G offers the more balanced geometry and the more directly usable public footprint data.

Source Code Availability:

The analysis code, tests, and generated-output workflow are available at https://github.com/ayberkdt/link_budget_analysis.

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